



CCS 249 Final Project Requirements

Requirements

For the Final Project in CCS 249, you and your group members are required to develop and submit an NLP System, which apply concepts discussed in the subject, from *text preprocessing* to get the features of the training corpus, to *developing models*, and *evaluating* said models.

The NLP system will consist of two components:

- a. A *Backend server* that will hold all of the NLP methods. The *Flask* framework for developing the backend is highly recommended the backend.
Flask For Beginners: <https://www.ffnext.io/blog/python-backend-with-flask-for-beginners>
- b. A *simple Frontend* that will accept user input, communicate with the backend via API, and display the output. Any frameworks/SDKs can be utilized.

Two documents are to be prepared:

- a. A software documentation encompassing the overview, introduction, features, technical aspects, and functional requirements of the NLP system.
- b. A short presentation is prepared to present the application's purpose, core features and a demonstration of the application's navigating and core features.

As for the groupings, a group can consist of up to 2-5 members.

The use of Python modules/libraries for preprocessing, model training, and evaluation is permitted as long as it is disclosed in the software documentation.

The use of third-party Language Models, Classifiers, and GPTs, especially APIs is prohibited. Models should be developed and trained by the group. *However, fine tuning a pretrained LLM model is allowed.*



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Finally, an assignment on the LMS will be created later for submitting the necessary files.

Specific requirements for submission:

1. Presentation

- a. A short PowerPoint/pdf presentation file to be presented at a scheduled date.
- b. Content of the presentation should include the following:
 - i. Problem statement and purpose of the app
 - ii. Identified the training corpora appropriate for the application.
 - iii. Text Preprocessing/Processing techniques explanation and implementation.
 - iv. NLP Task identified, explained and its implementation.
 - v. Model Training and Development
 - vi. Evaluation of the model
- c. Presentation time should be 5-10 minutes long
- d. A 5-10 minute Q&A session

2. *A software documentation listing the technical aspects of the developed NLP system.*

- a. The documentation must contain the following sections:
 - i. Overview and Introduction
 1. App Name
 2. Purpose
 3. Technology Stack (Frameworks, SDKs, modules/packages utilized) along with their versions
 - ii. Functional Requirements
 1. Core features with Descriptions
 2. Specify libraries/packages.
 - iii. Future Enhancements and Roadmap
 1. Known Limitations



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2. Planned Features and Improvements

3. *GitHub submission of the project source code and folders.*

- a. A GitHub repository is going to be created within our organization.
- b. The team leaders will fork the repository, collaborate with their team members.
- c. Once the project is done, the team leaders will create a PR to merge their changes to the main repository.

4. Self and Peer assessment

- a. A Self and Peer Assessment template will be distributed to everyone a day after your project presentation.
- b. This will factor in the final project score, as stated in the Syllabus:

Project Score Calculation: $(PROJECT\ SCORE\ RAW * 50\%) + (SELF\ EVAL\ SCORE\ RAW * 25\%) + (PEER\ EVAL\ SCORE\ RAW * 25\%)$